SHAMANTH S

shamanthsheni@gmail.com | +91 8891429533 | LinkedIn | GitHub

SUMMARY

AI/ML undergraduate with hands-on experience in **Machine Learning, Deep Learning, NLP** and **Computer Vision**. Worked on research projects involving **computational pathology, crowd counting** and AI-based research analysis. Currently working as a research intern on **nuclear pleomorphism analysis** in histopathological images. Interested in AI research and practical machine learning applications.

EDUCATION

B.E in Artificial Intelligence and Machine Learning — **CGPA – 7.7**
Jyothy Institute of Technology (VTU), Bengaluru — **June 2027 ***
Coursework: Mathematics, Machine Learning, Artificial Intelligence, Data Science, Software Engineering

Pre-University (PCMB) — **Aggregate: 90%**
Ambika Padavipoorva Vidyalaya, Puttur — **March 2023**

TECHNICAL SKILLS

- **Programming Languages:** Python, C, MySQL, MongoDB
- **ML & DL:** Machine Learning, PyTorch, CNNs, Transfer Learning and Attention Mechanisms
- **NLP & AI:** Sentence-BERT, BERTopic and Embeddings
- **Computer Vision:** Image Processing and Computational Pathology
- **DevOps:** AWS, Docker, GitHub and Linux

EXPERIENCE

Research Intern – Computational Engineering Lab, CIIRC — **May 2026 – Present**

- Working on **nuclear pleomorphism analysis** and **automated Nottingham grading** using histopathological images.
- Experimented with **SKDenseNet, U-Net, CellViT** and **StarDist** for nucleus detection, segmentation and classification.
- Analyzing **tumor nuclear morphology** for quantitative pleomorphism assessment.
- Studying research papers and testing different approaches used in **computational pathology**.

PROJECTS

MSA-CSRNet – Multi-Scale Crowd Counting

- Reimplemented **CSRNet** in **PyTorch** using **VGG16** and dilated convolutions for density-based crowd counting on **ShanghaiTech Part A**.
- Added **multi-scale feature fusion** and **SE attention** to improve feature representation. Reduced **MAE from 73.78 to 62.81** (14.9% improvement) with a **correlation of 0.957**.

Strategic Research Miner – AI Research Intelligence Platform

- Built an AI pipeline to analyze **80,000+ research papers** using **Sentence-BERT** and **BERTopic** to identify research trends and potential gaps.
- Added **LSTM-based forecasting** with MAE of approximately **0.077** along with strategic scoring and hypothesis generation. Validated results using **2024–2025 arXiv data**.

BharathBrief – AI Problem Intelligence Platform

- Built an automated pipeline using **Google Gemini** to collect information from **250+ news and public sources** and convert it into problem briefs.
- Developed a **Next.js + Supabase** platform that processes **1,000+ extracted problems**. Added **GitHub Actions** for daily pipeline execution.

CERTIFICATES

- **Oracle AI Foundations** – Oracle (Oct 2025) – Certificate
- **AWS Academy Foundations** – AWS (Oct 2025) – Certificate
- **C for Everyone** – University of California, Santa Cruz (Mar 2025) – Certificate